

Summary and Recommendations

Executive Summary – Customer Churn Analysis

This exploratory data analysis (EDA) examines customer churn patterns in a telecom dataset using various visualizations such as count plots, stacked bar charts, and distribution graphs. The goal is to identify key drivers of churn and provide actionable insights for improving customer retention.

♦ Overall Churn Rate

The dataset shows that approximately **26.5% of customers have churned**, while **73.5% have retained the service**. This indicates that more than one-quarter of the customer base is at risk, making churn reduction a critical business priority.

♦ Demographic Insights

- **Senior Citizens:**
 - Around **41–42% of senior citizens churn**, compared to only **~23–24% of non-senior customers**.
 - This clearly indicates that senior citizens are **nearly 2x more likely to churn**.
 - **Gender:**
 - Churn distribution is almost equal:
 - Male: ~26%
 - Female: ~27%
 - This suggests that **gender has no significant impact on churn behavior**.
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♦ Tenure Analysis

- Customers with **low tenure (0–3 months)** show the highest churn rates (**~50% or more** in some cases).
 - Customers with **long tenure (2+ years)** have significantly lower churn (**<10–15%**).
 - This highlights that **early-stage customers are the most vulnerable**, and improving onboarding can reduce churn.
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♦ Service-Based Insights

Analysis of service-related features shows strong patterns:

- **Internet Service:**
 - Fiber optic users show **higher churn (~40%+)** compared to DSL users (~20%).
 - **Online Security / Tech Support:**
 - Customers **without these services** have churn rates of **~40–45%**.
 - Customers **with these services** churn much less (**~15–20%**).
 - Indicates these services are **key retention drivers**.
 - **Streaming Services (TV, Movies):**
 - Slightly higher churn among users, possibly due to **higher costs or expectations**.
 - **Phone Service & Multiple Lines:**
 - Minimal impact on churn; distribution remains fairly balanced.
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◆ Key Behavioral Patterns

- Customers with **fewer value-added services** are more likely to churn
 - **High monthly charges** (observed in charts) are associated with increased churn
 - **New customers + high cost + lack of support services = highest churn risk group**
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Key Insights (Quick View)

- **26.5% churn rate** → significant retention issue
- **Senior citizens churn ~2x more** than others
- **New customers (<3 months) are highest risk**
- **Support services reduce churn by ~50%**
- **Fiber optic users show higher churn (~40%+)**